A2: Weight decay and phase transitions

Worked solutions

Training dynamics, a self-study course · written by Claude for Fan Pu

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`reference numbers: solutions/results.json · figures: solutions/figures/ · equation numbers (n) refer to the handout`

Every derivation here is complete; read only as far as you need. "Measured" numbers come from running each check script with the reference implementations (`--solutions`) on CPU; your numbers should agree within the tolerances stated in each problem. Diff yours with `uv run python -m tdyn.results a2-weight-decay/out/results.json a2-weight-decay/solutions/results.json`. Equation numbers such as (1) refer to the handout; this document numbers its own equations (S1), (S2), … Every closed form below was checked symbolically (sympy) and numerically before it was written down.

1 Problem (`balance`): Weight decay balances the layers

(a) With $\nabla \ell(ab) = \ell'(ab)(b, a)$, the loss part cancels as in A1 and only weight decay is left:

$$a\,\partial_a L_\gamma - b\,\partial_b L_\gamma = a(\ell' b + 2\gamma a) - b(\ell' a + 2\gamma b) = 2\gamma(a^2 - b^2) = 2\gamma c. \tag{S1}$$

At a stationary point the left side is zero, so for $\gamma > 0$ every stationary point has $a^2 = b^2$, whatever ℓ and the data. Along the flow $\dot c = 2a\dot a - 2b\dot b = -2\big(a\partial_a L_\gamma - b\partial_b L_\gamma\big) = -4\gamma c$, so $c(t) = c_0 e^{-4\gamma t}$ (A1 Problem (`broken`) (a)). The conserved quantity of A1 becomes a decaying one, and its only fixed value is 0.

(b) With $G = \nabla_W L$ at $W = W_2 W_1$, the gradients are $\nabla_{W_1} = W_2^\top G + 2\gamma W_1$ and $\nabla_{W_2} = G W_1^\top + 2\gamma W_2$. The G terms cancel in $\dot D$ exactly as in A1 Problem (`matrix-law`) (b); the weight-decay terms give $\dot W_1 W_1^\top + W_1 \dot W_1{}^\top \supset -4\gamma W_1 W_1^\top$ and $-\big(\dot W_2{}^\top W_2 + W_2^\top \dot W_2\big) \supset +4\gamma W_2^\top W_2$. So

$$\dot D = -4\gamma D, \quad D(t) = D(0) e^{-4\gamma t}. \tag{S2}$$

At any stationary point $D = 0$: $W_1 W_1^\top = W_2^\top W_2$. The (j, j) entry reads $\|W_1[j,:]\|^2 = \|W_2[:, j]\|^2$: each hidden neuron has the same norm coming in as going out. This is Ziyin et al.'s Lemma 1 with $\gamma_u = \gamma_w$ (with unequal decay strengths the flow gives $\dot D = -4(\gamma_w W_1 W_1^\top - \gamma_u W_2^\top W_2)$ and the stationary relation becomes their $\gamma_u U_j^2 = \gamma_w \|W_{j:}\|^2$). Here it holds at *every* stationary point, not only the global minimum. The off-diagonal entries say more: the hidden representations of the two layers have equal Gram matrices.

(c) For $f = abcx$, $\partial_a L_\gamma = bc\,\ell' + 2\gamma a$ and so on, and $a\partial_a L_\gamma - b\partial_b L_\gamma = 2\gamma(a^2 - b^2)$. Hence $a^2 - b^2$, $b^2 - c^2$ (and $a^2 - c^2$) each decay as $e^{-4\gamma t}$, and every stationary point has $a^2 = b^2 = c^2$.

(d) Expected. Maximum relative error of $D(t)$ on $[0, 20]$: 2.2×10^{-11} ($\gamma = 0.05$) and 1.9×10^{-10} ($\gamma = 0.2$); depth-3 differences 1.1×10^{-10}. $\max \|a_\infty| - |b_\infty\|$ over 200 starts: 1.5 at $\gamma = 0$ (endpoints spread along $ab = 1$ between $(0.5, 2)$ and $(2, 0.5)$) and 7.8×10^{-16} at $\gamma = 0.1$ (all at $(\pm 0.894, \pm 0.894)$). Matrix network at $t = 200$: $\|D\|_F = 3.2 \times 10^{-16}$, per-neuron gap 1.1×10^{-16}.

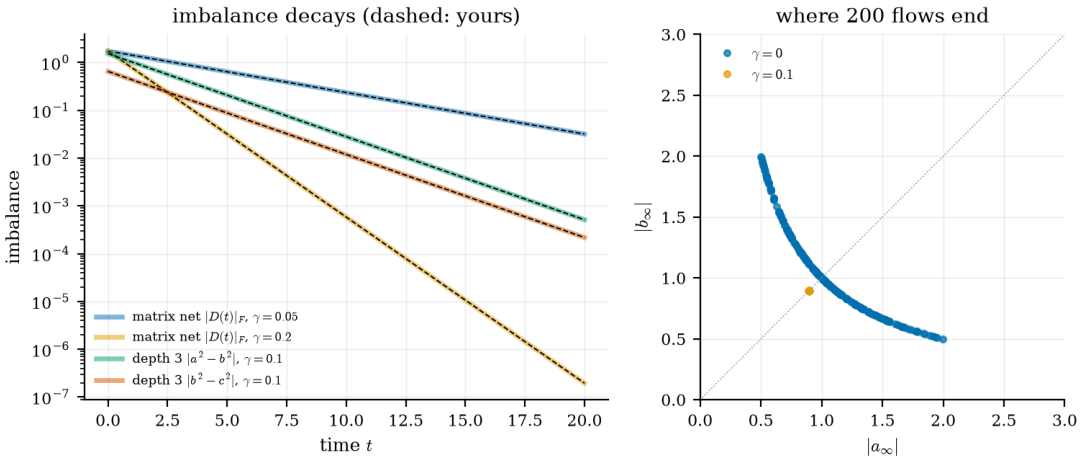


Figure 1: `balance.png`. Left: imbalances decaying at rate 4γ (dashed: the formula). Right: where 200 flows end without (blue) and with (orange) weight decay.

2 Problem (`stationary`): Stationary points and the bifurcation at one half

(a) By (balance) (a), a stationary point has $b = a$ or $b = -a$.

- $b = a$: $\partial_a L_\gamma = ar + 2\gamma a = a(a^2 - 1 + 2\gamma) = 0$, so $a = 0$ or $a^2 = 1 - 2\gamma$, possible only for $\gamma < \frac{1}{2}$.
- $b = -a$: $r = -a^2 - 1$ and $\partial_a L_\gamma = br + 2\gamma a = a(a^2 + 1 + 2\gamma)$, zero only at $a = 0$.

So the stationary points are the origin for every γ, and $\pm(u, u)$ with $u = \sqrt{1 - 2\gamma}$ for $\gamma < \frac{1}{2}$. From any start off the line $a = -b$ the flow ends at one of $\pm(u, u)$ for $\gamma < \frac{1}{2}$ (the origin is a saddle, below) and at the origin for $\gamma > \frac{1}{2}$:

$$p_\infty(\gamma) = \begin{cases} 1 - 2\gamma & \gamma < \frac{1}{2} \\ 0 & \gamma \geq \frac{1}{2}. \end{cases} \tag{S3}$$

The Hessian of (1) is

$$H(a, b) = \begin{pmatrix} b^2 + 2\gamma & 2ab - 1 \\ 2ab - 1 & a^2 + 2\gamma \end{pmatrix}. \tag{S4}$$

At the origin $H = \left(\begin{smallmatrix} 2\gamma & -1 \\ -1 & 2\gamma \end{smallmatrix}\right)$: eigenvalue $2\gamma - 1$ on $(1, 1)$ and $2\gamma + 1$ on $(1, -1)$. At $\pm(u, u)$, $a^2 = b^2 = ab = 1 - 2\gamma$, so $H = \left(\begin{smallmatrix} 1 & 1-4\gamma \\ 1-4\gamma & 1 \end{smallmatrix}\right)$: eigenvalue $2 - 4\gamma$ on $(1, 1)$ and 4γ on $(1, -1)$. Classification:

- $\gamma < \frac{1}{2}$: origin a saddle ($2\gamma - 1 < 0 < 2\gamma + 1$); $\pm(u, u)$ minima ($4\gamma > 0$, $2 - 4\gamma > 0$).
- $\gamma = \frac{1}{2}$: only the origin, degenerate (eigenvalues 0 and 2). It is still the global minimum: on $a = b$, $L_{1/2} = \frac{1}{2}(u^2 - 1)^2 + u^2 = \frac{1}{2} + u^4/2$, flat to fourth order.
- $\gamma > \frac{1}{2}$: the origin is the only stationary point and a minimum.

As $\gamma \uparrow \frac{1}{2}$ the two minima slide along $a = b$ into the origin, and their eigenvalue $2 - 4\gamma$ along the line goes to zero together with the origin's $2\gamma - 1$. This is a *supercritical pitchfork bifurcation*: one fixed point (the origin) changes stability while two symmetric ones are born from it. The order parameter $p_\infty = 1 - 2\gamma = 2\left(\frac{1}{2} - \gamma\right)$ vanishes linearly; in terms of the weights, $u = \sqrt{2\left(\frac{1}{2} - \gamma\right)}$ vanishes like a square root, the mean-field exponent of a magnet.

(b) Reference: `solutions/stationary_solution.py`.

(c) Expected. Maximum $|ab - p_\infty|$ over 79 values of γ: 9.8×10^{-9} (the run at $\gamma = 0.499$ is the slowest). Autograd eigenvalues equal to the formulas to 2.2×10^{-16}; e.g. at $\gamma = 0.45$: origin $(-0.1, 1.9)$, minima $(0.2, 1.8)$. Log-log slope of p_∞ against $\frac{1}{2} - \gamma$: 1.0000. Against the ridge curve $1/(1+2\gamma)$ of the example, depth 2 shrinks faster and hits zero.

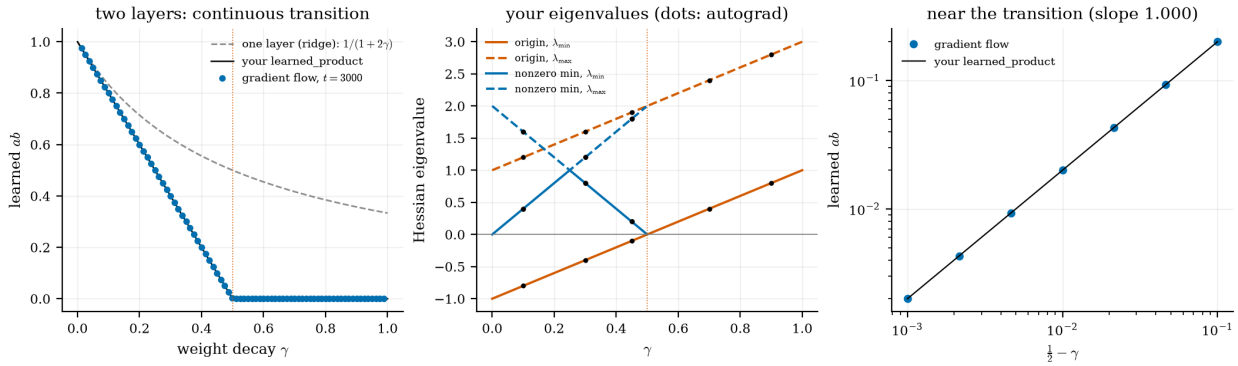


Figure 2: `bifurcation.png`. Left: learned ab against γ. Middle: Hessian eigenvalues at the origin and the minima. Right: the order parameter near $\gamma = \frac{1}{2}$.

3 Problem (`slowing`): Critical slowing down

(a) On $a = b = u$, (2) gives $\dot{u} = -u(u^2 - 1 + 2\gamma)$, and $\dot{p} = 2u\dot{u} = 2p(s_{\text{eff}} - p)$ with

$$s_{\text{eff}} = 1 - 2\gamma. \tag{S5}$$

Weight decay lowers the effective target; above $\gamma = \frac{1}{2}$ the target is negative and the logistic equation drives p to 0.

(b) The flow converges to the minimum for $\gamma < \frac{1}{2}$ and to the origin for $\gamma > \frac{1}{2}$, so with (stationary) (a):

$$\tau(\gamma) = \begin{cases} 1/(4\gamma) & \text{if } 0 < \gamma \le \frac{1}{4} \\ 1/(2 - 4\gamma) = 1/\big(4(\frac{1}{2} - \gamma)\big) & \text{if } \frac{1}{4} \le \gamma < \frac{1}{2} \\ 1/(2\gamma - 1) = 1/\big(2(\gamma - \frac{1}{2})\big) & \text{if } \gamma > \frac{1}{2}. \end{cases} \tag{S6}$$

The slow direction is $(1, -1)$ on the first piece and $(1, 1)$ on the other two. So $C_- = \frac{1}{4}$, $C_+ = \frac{1}{2}$, and $C_+/C_- = 2$: at equal distance from the critical point, relaxation is twice as slow on the zero side. (The same factor 2 appears in the Landau theory of a magnet, for the same reason: below the transition the restoring curvature comes from both the quadratic and the quartic term.) The other divergence is $\gamma \to 0$, along $(1, -1)$: that is the imbalance c, which decays only at rate 4γ (balance (a)); at $\gamma = 0$ it is A1's conserved quantity and the minima form a flat valley.

(c) At $\gamma = \frac{1}{2}$, $\dot{p} = -2p^2$, so $1/p = 1/p_0 + 2t$ and

$$p(t) = \frac{p_0}{1 + 2p_0 t}, \quad \|\theta(t)\| = \sqrt{2p} \approx t^{-1/2}. \tag{S7}$$

Convergence is algebraic, not exponential: the linear term vanished and only the cubic force $-u^3$ is left.

(d) The distance on the line is $\sqrt{2}\,|u - u^*|$ with $u^* = \sqrt{s_{\text{eff}}}$ (or 0 for $\gamma \ge \frac{1}{2}$). The flow reaches distance ε at $u_\varepsilon = u^* + \text{sign}(u_0 - u^*)\,\varepsilon/\sqrt{2}$, i.e. $p_\varepsilon = u_\varepsilon^2$. Inverting A1's logistic solution with target $s = s_{\text{eff}}$,

$$t_\varepsilon = \frac{1}{2s} \ln \frac{p_\varepsilon (s - p_0)}{p_0 (s - p_\varepsilon)} \quad (s \neq 0), \quad t_\varepsilon = \frac{1}{2}\left(\frac{1}{p_\varepsilon} - \frac{1}{p_0}\right) \quad (s = 0), \tag{S8}$$

and $t_\varepsilon = 0$ if $\sqrt{2}\,|u_0 - u^*| \le \varepsilon$. For $s < 0$ both $s - p_0$ and $s - p_\varepsilon$ are negative, so the logarithm is real. The slope is not exactly -1: $t_\varepsilon \approx \tau \ln(K/\varepsilon)$ and K depends on γ. Below, $s - p_\varepsilon \approx -\sqrt{2s}\,\varepsilon$ gives $K \propto \sqrt{s}$;

above, $p_\varepsilon = \varepsilon^2/2$ gives $t_\varepsilon \approx \frac{1}{2|s|}\ln(2\,|s|\,p_0/((p_0+|s|)\varepsilon^2))$. In both, $d\ln t_\varepsilon / d\ln|\gamma - \frac{1}{2}| = -1 + O(1/\ln(K/\varepsilon))$, slightly shallower than -1. The late-time ratio used in the check removes K.

(e) Reference: `solutions/slowing_solution.py`.

(f) Expected. Measured over predicted τ: within 2.6×10^{-4} below and 1.4×10^{-6} above; slopes -0.9999 and -1.0000; $\tau\,|\gamma - \frac{1}{2}| = 0.2500$ and 0.5000, ratio 2.0002. t_ε ($\varepsilon = 10^{-4}$, $u_0 = 0.5$) matches Equation (S8) to 4×10^{-9} relative; its slopes are -0.962 (below) and -0.940 (above). At $\gamma = \frac{1}{2}$: $p(t)$ matches to 1.5×10^{-10}, late-time exponent -0.4998.

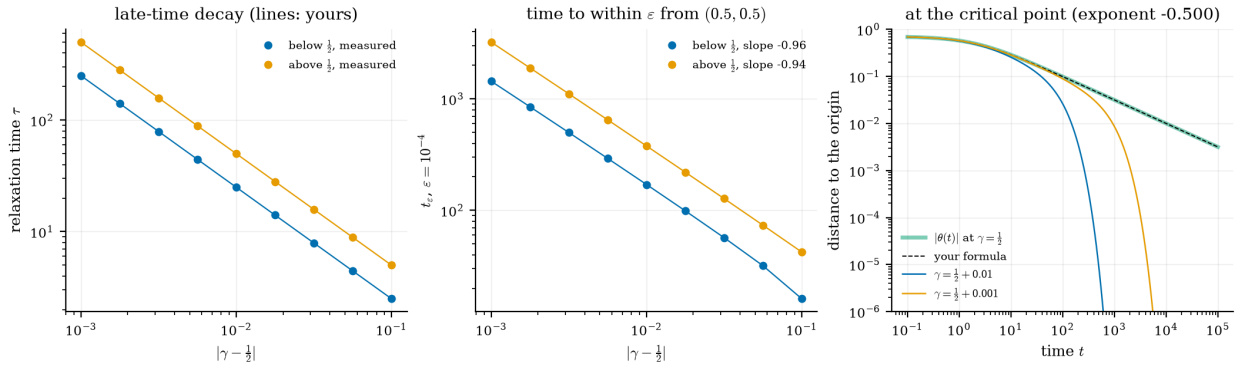


Figure 3: `slowing.png`. Left: relaxation time on both sides (the upper line is the zero side). Middle: time to within 10^{-4}. Right: algebraic decay at $\gamma = \frac{1}{2}$ against exponential decay just above it.

4 Problem (`depth3`): The jump at depth 3

(a) $\frac{1}{2}(abc - s)^2 = \frac{1}{2}s^2 - s\,abc + \frac{1}{2}a^2b^2c^2$. Apart from the constant, every term has degree 3 or 6, so all second derivatives of the data term vanish at the origin, and

$$\nabla^2 L_\gamma(0,0,0) = 2\gamma I. \tag{S9}$$

The origin is a strict local minimum for every $\gamma > 0$. At depth 2 the term $-s\,ab$ is quadratic and gives the off-diagonal $-s$ that, for $2\gamma < s$, makes the origin a saddle. At depth 3 the signal enters only at cubic order, so near zero weight decay always wins (Ziyin et al., Proposition 2).

(b) Multiplying the components of $\nabla L_\gamma = 0$ by a, b, c: $pr + 2\gamma a^2 = pr + 2\gamma b^2 = pr + 2\gamma c^2 = 0$, so $a^2 = b^2 = c^2 = u^2$ and $pr = -2\gamma u^2 < 0$ at a nonzero point. With $r = p - s$ that forces $0 < p < s$, hence $p = abc = u^3 > 0$ (an even number of negative weights). If one weight is zero, the others' equations reduce to $2\gamma b = 0$ and so on, which gives the origin. Then $pr = -2\gamma u^2$ reads $u^3(u^3 - s) = -2\gamma u^2$, i.e.

$$u^4 - su + 2\gamma = 0 \quad\Leftrightarrow\quad \gamma = g(u) = u\frac{s - u^3}{2}. \tag{S10}$$

$g(0) = 0$, g rises to a maximum and falls to 0 at $u = s^{1/3}$: two positive roots for $0 < \gamma < \max g$, none above. $g'(u) = (s - 4u^3)/2 = 0$ at $(u^*)^3 = s/4$, so

$$\gamma^*(s) = g(u^*) = (s/4)^{1/3} \cdot \frac{3s/4}{2} = \frac{3}{8}\,4^{-1/3}s^{4/3}, \quad \gamma^*(1) = \frac{3 \cdot 2^{1/3}}{16} = 0.23624, \tag{S11}$$

and the product just below γ^* is $p = (u^*)^3 = s/4$. Followed from $\gamma = 0$ (where $p = s$), the learned product decreases to $s/4$ and then jumps to 0: a discontinuous transition.

(c) At (u,u,u): $\partial_a^2 L_\gamma = (bc)^2 + 2\gamma = u^4 + 2\gamma$ and $\partial_a\partial_b L_\gamma = cr + abc^2 = u(u^3 - s) + u^4$. Eliminating $2\gamma = su - u^4$ with Equation (S10): $\alpha = su$, $\beta = 2u^4 - su$. The eigenvalues of $\alpha I + \beta(\mathbf{1}\mathbf{1}^\top - I)$ are $\alpha + 2\beta$ on $(1,1,1)$ and $\alpha - \beta$ (twice) across it:

$$\lambda_{\parallel} = u(4u^3 - s), \quad \lambda_{\perp} = 2u(s - u^3) = 4\gamma > 0. \tag{S12}$$

So the larger root ($u^3 > s/4$) is a minimum and the smaller root ($u^3 < s/4$) is a saddle with one unstable direction, along the ray. At γ^* the two meet at $u^3 = s/4$ where $\lambda_{\parallel} = 0$ and annihilate: a *saddle-node bifurcation*.

(d) With $q = u^3$ and Equation (S10), $3\gamma u^2 = \frac{3}{2}q(s-q)$ and

$$L_\gamma = \frac{1}{2}(q-s)^2 + \frac{3}{2}q(s-q) = \frac{s^2}{2} + \frac{qs}{2} - q^2. \tag{S13}$$

This equals $s^2/2$ at $q = s/2$, so $u_c = (s/2)^{1/3}$ and

$$\gamma_c(s) = g(u_c) = \frac{s}{4}\left(\frac{s}{2}\right)^{1/3}, \quad \gamma_c(1) = 2^{-1/3}/4 = 0.19843. \tag{S14}$$

On the minimum branch q decreases with γ, so for $\gamma_c < \gamma < \gamma^*$ we have $q < s/2$ and $L > s^2/2$: the origin is the global minimum and the learned solution is only *metastable*. Gradient flow does not see γ_c; it stays on the nonzero minimum until that minimum disappears at γ^*.

(e) Expected prediction. Going up, *abc* decreases smoothly to 1/4 and then drops to 0 at γ^*. Going down, the depth-3 net never learns again, down to $\gamma \to 0$: the origin is a minimum for every $\gamma > 0$, and a 10^{-3} kick is far inside its basin (radius about 2γ, Problem (`basins`)) except at the very smallest γ. The depth-2 net retraces its curve exactly: its origin is a saddle below 1/2 and any kick escapes.

(f) Reference: `solutions/depth3_solution.py`.

(g) Expected. Hessian at the origin diag(0.02, 0.02, 0.02), (0.2, 0.2, 0.2), (1, 1, 1) with zero off-diagonals. Learned *abc* on the larger root cubed to 2.3×10^{-10} away from γ^*. Bisection γ^* against the formula: 0.0372048 vs 0.0372047 ($s = 0.25$), 0.0937501 vs 0.0937500, 0.2362353 vs 0.2362352, 0.5952752 vs 0.5952754 ($s = 2$), all within 2×10^{-6} relative; products just before the jump 0.0625, 0.1250, 0.2503, 0.5009 against $s/4$ (the last bisection point is slightly below γ^*). Log-log slope of γ^* in s: 1.3333 = 4/3. Ziyin et al.'s Proposition 3 (their (15)) has $\|\mathbb{E}[xy]\|^{1+(D-1)/(D+1)}$, which is $s^{4/3}$ at $D = 2$: the same power. At depth 2 the threshold is $s/2$, linear in s, so weak signals lose much sooner at depth 3. Sweep: depth 2 gives $p =$ 0.800, 0.400 at $\gamma = 0.1, 0.3$ in both directions; depth 3 gives 0.783 and 0 going up and 0, 0 coming down.

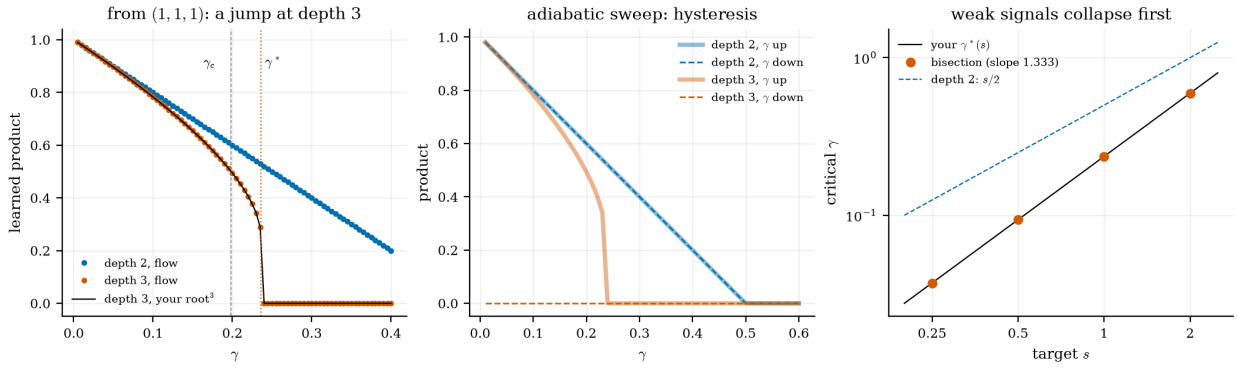


Figure 4: `depth3.png`. Left: learned product from (1, 1, 1). Middle: adiabatic sweeps; the depth-3 loop never closes. Right: $\gamma^*(s)$ against the depth-2 threshold $s/2$.

5 Problem (`basins`): Basins of the zero solution

(a) L_γ is symmetric under permutations of (a, b, c), so at a point with $a = b = c$ the three gradient components are equal and the flow stays on the ray. There

$$\dot{u} = F(u) = -u(u^4 - su + 2\gamma). \tag{S15}$$

The bracket equals $2\gamma > 0$ at $u = 0$, is positive for all $u < 0$ (all three terms are), and for $\gamma < \gamma^*$ is negative exactly between the roots $u_s < u_+$ of Equation (S10). Phase line: $F > 0$ for $u < 0$, $F < 0$ on $(0, u_s)$, $F > 0$ on (u_s, u_+), $F < 0$ above u_+. Endpoint rule: $u_0 \le 0 \to 0$; $0 < u_0 < u_s \to 0$; $u_0 = u_s$ stays; $u_0 > u_s \to u_+$. For $\gamma > \gamma^*$ every start goes to 0. For small γ the root u_s is small, the u^4 term is negligible, and

$$u_s \approx \frac{2\gamma}{s} \quad \left(\text{next term: } u_s = \frac{2\gamma}{s} + \frac{(2\gamma)^4}{s^5} + \ldots\right). \tag{S16}$$

The zero basin's radius grows linearly in the weight decay and inversely in the signal, the $\gamma/\|\mathbb{E}[xy]\|$ scaling of Ziyin et al.'s bound (16) at $D = 2$.

(b) Swapping a and b is a symmetry, so the plane $a = b$ is invariant. In it the basin boundary is the stable manifold of the balanced saddle (u_s, u_s, u_s): in 3-D it is a surface (the saddle has two stable directions, eigenvalue -4γ of the flow's Jacobian, and one unstable), and the plane cuts it in a curve through the saddle tangent to $(1, 1, -2)$. Flipping the signs of two weights, $(a, b, c) \to (-a, -b, c)$, leaves abc and the loss unchanged and maps the plane to itself by $(x, y) \to (-x, y)$; it takes the saddle to $(-u_s, -u_s, u_s)$, whose stable manifold is the mirrored boundary.

(c) Reference: `solutions/basins_solution.py`.

(d) Expected. Balanced starts: maximum error 1.3×10^{-11}, measured edge between 0.200 and 0.210 around $u_s = 0.20165$ ($\gamma = 0.1$); 67% of the plane's grid learns. Gaussian starts at $\gamma = 0.03$: $u_s = 0.0600$ for $s = 1$ and 0.1513 for $s = 0.4$; the learning fraction is 0.44 at $\sigma = 0.100$ and 0.65 at 0.133 for $s = 1$, and 0.40 at 0.237, 0.57 at 0.316 for $s = 0.4$, so $\sigma_{1/2} \approx 0.11$ and 0.28: about $1.8u_s$ in both cases. Exact over small-γ radius: 1.0000 up to $\gamma = 0.01$, 1.0002 at 0.03, 1.0083 at 0.1.

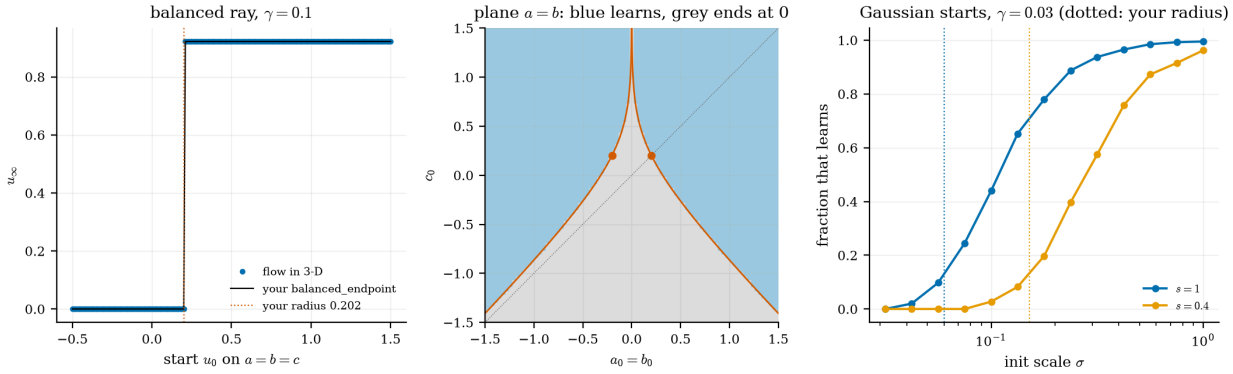


Figure 5: `basins.png`. Left: endpoints on the balanced ray. Middle: the plane $a = b$, blue learns, grey ends at zero, with the saddles' stable manifolds. Right: fraction of Gaussian starts that learn, with the basin radius dotted.

6 Problem (`finite-step`): The phase diagram of GD with weight decay

(a) Near a fixed point, $\theta_{k+1} - \theta^* \approx (I - \eta H)(\theta_k - \theta^*)$, which contracts iff $|1 - \eta\lambda| < 1$, i.e. $0 < \lambda < 2/\eta$, for every eigenvalue. With (stationary) (a):

$$\eta_{\max}(\gamma) = \begin{cases} 2/(2 - 4\gamma) = 1/(1 - 2\gamma) & \text{if } 0 < \gamma \le \frac{1}{4} \\ 2/(4\gamma) = 1/(2\gamma) & \text{if } \frac{1}{4} \le \gamma < \frac{1}{2} \\ 2/(2\gamma + 1) & \text{if } \gamma > \frac{1}{2}. \end{cases} \tag{S17}$$

The direction that loses stability is $(1,1)$ at the minimum on the first piece, $(1,-1)$ at the minimum on the second, and $(1,-1)$ at the origin on the third. The maximum is $\eta_{\max}\left(\frac{1}{4}\right) = 2$, where the two eigenvalues at the minimum are equal (both 1). The limits at $\gamma \to \frac{1}{2}$ are 1 from both sides, so the boundary is continuous across the transition, and $\eta_{\max} \to 1$ as $\gamma \to 0$ (the $(1,1)$ minimum of the plain anchor has sharpness $a^2 + b^2 = 2$, A3). For $\gamma < \frac{1}{2}$ the origin cannot rescue GD: its eigenvalue $2\gamma - 1 < 0$ makes $|1 - \eta\lambda| > 1$ for every η.

(b) Expected prediction. A band of bounded oscillation only for $\gamma < \frac{1}{4}$. There the unstable direction is $(1,1)$, along the product, where the cubic nonlinearity folds the iterates back (the same mechanism as edge-of-stability oscillations in A3). For $\gamma > \frac{1}{4}$ the unstable direction is the imbalance: at the minimum $r = -2\gamma$ and A1's factor is $c_{k+1}/c_k = (1 - 2\eta\gamma)^2 - \eta^2 r^2 = 1 - 4\eta\gamma$, which drops below -1 exactly at $\eta = 1/(2\gamma)$. Then $|c|$ grows, $a^2 + b^2 \geq |c|$ grows, the curvature grows with it, and nothing saturates: divergence. Likewise past $2/(2\gamma + 1)$ at the origin.

(c) On $a = b = u$ the GD map is $u \mapsto u - \eta u(u^2 - s)$, $s = s_{\text{eff}}$. Its derivative at $u^* = \sqrt{s}$ is $1 - \eta\left(3(u^*)^2 - s\right) = 1 - 2\eta s$, so the fixed point is stable iff $\eta < 1/s$: oscillation onset $\eta = 1/(1 - 2\gamma)$ (the first piece of Equation (S17), as it should be). With $u = kv$ and $\eta k^2 = \mu = 1 + \eta s$, the map is $v \mapsto \mu(v - v^3)$. At $v_e = \sqrt{1 + 1/\mu}$, $\mu(v_e - v_e^3) = \mu v_e(1 - v_e^2) = -v_e$, so the endpoints of $[-v_e, v_e]$ map to each other, and outside it $|f(v)| > |v|$ grows without bound. The interval maps into itself iff the local maximum $f\left(1/\sqrt{3}\right) = 2\mu/\left(3\sqrt{3}\right)$ is at most v_e:

$$\frac{4\mu^2}{27} \leq 1 + \frac{1}{\mu} \quad \Leftrightarrow \quad 4\mu^3 - 27\mu - 27 = (\mu - 3)(2\mu + 3)^2 \leq 0 \quad \Leftrightarrow \quad \mu \leq 3. \tag{S18}$$

For $\mu > 3$ the top of the map pokes out of the interval, and almost every orbit eventually lands outside and escapes. With $\mu = 1 + \eta s$: divergence onset $\eta = 2/s = 2/(1 - 2\gamma)$. Between $1/s$ and $2/s$ the map goes through period doubling and chaos, the same route as the logistic map.

(d) Reference: `solutions/finite_step_solution.py`.

(e) Expected. From $(0.6, 0.4)$: 2746 cells converge to the nonzero minimum, 1594 to zero, 818 oscillate and 4842 diverge; agreement with `predicted_phase` is 1.0000 on the 9739 cells more than 3% from the boundary and 0.9998 overall. The two disagreeing cells ($\gamma = 0.525$, $\eta = 0.975$ and $\gamma = 0.675$, $\eta = 0.850$) sit within 0.3% below $\eta_{\max}$ and diverge: the origin is linearly stable there, but its basin shrinks to nothing at the boundary and $(0.6, 0.4)$ is outside it. Oscillating cells occur only for $\gamma \in [0.005, 0.245]$, below $1/4$. Their upper edge (between $\eta \approx 2$ and 2.5) is ragged: it is set by when a chaotic orbit lands outside the trapping region, which the linear analysis does not predict, and from an unbalanced start it lies below the balanced line's $2/(1 - 2\gamma)$ for larger γ (the imbalance is not stable along a chaotic orbit). From $(0.45, 0.45)$: both onsets match (c) within 0.8% and 0.4%, the η grid step. The post's caption calls the stable-solution boundaries analytic and the oscillation and divergence regions numerical; here the stability boundaries are Equation (S17) and the line $\gamma = \frac{1}{2}$, and on the balanced line the divergence edge is analytic too.

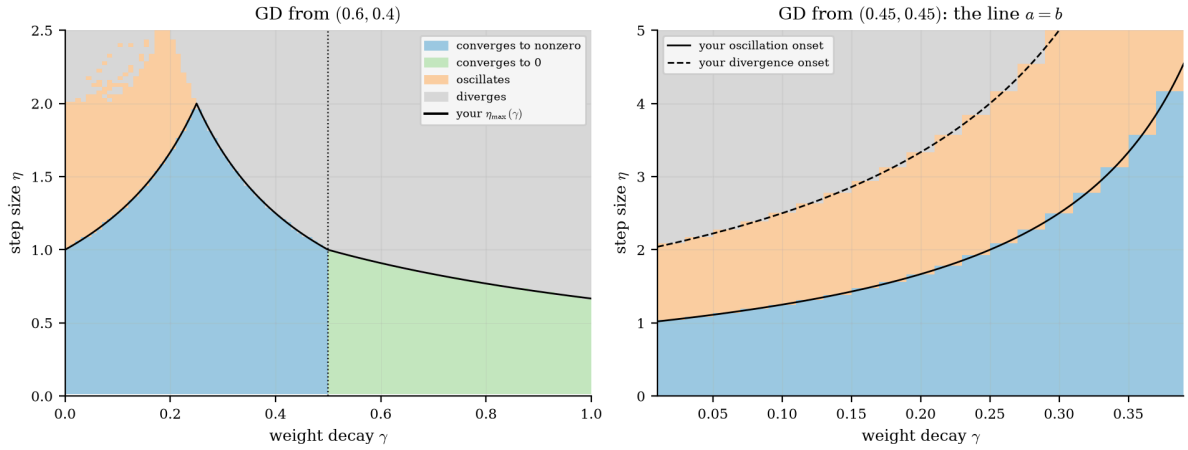


Figure 6: `finite_step.png`. Left: the phase diagram of GD from $(0.6, 0.4)$ with $\eta_{\max}(\gamma)$. Right: the balanced line, with the oscillation and divergence onsets.

7 Problem (`ziyin`): Exact solutions of a deep linear network

(a) $\mathbb{E}_\varepsilon\left[\frac{1}{2}(a\varepsilon b - 1)^2\right] = \frac{1}{2}((1+\sigma^2)a^2b^2 - 2ab + 1) = \frac{1}{2}(ab-1)^2 + \frac{\sigma^2}{2}a^2b^2$: the noise is a penalty on the *product* (on the function), not on the weights. Using balance, $a^2 + b^2 \geq 2|ab|$ with equality at the optimum, so minimize $\frac{1}{2}(p-1)^2 + \sigma^2 p^2/2 + 2\gamma p$ over $p \geq 0$: $p_\infty = (1-2\gamma)/(1+\sigma^2)$ for $\gamma < \frac{1}{2}$ and 0 above. The critical γ stays at $\frac{1}{2}$ because the noise term is quartic in the weights and does not touch the Hessian at the origin; it only shrinks the learned product. This is their remark in §5 that the two-layer existence condition $\|\mathbb{E}[xy]\|^2 > \gamma_u \gamma_w$ is independent of the noise (and of width).

(b) At the origin the Hessian only sees the quadratic part of the loss. With one hidden layer, $-s\,ab$ is quadratic and gives off-diagonal entries $-s$; when the signal beats the weight decay ($s > 2\gamma$ here) they make the origin a saddle, otherwise it is the global minimum (Theorem 1). With two or more hidden layers the signal term $-s\,abc$ is cubic or higher, the Hessian is $2\gamma I$, and the origin is a local minimum for every $\gamma > 0$ (Proposition 2), a bad one whenever a nonzero solution has lower loss ($\gamma < \gamma_c$ here).

(c) The Gaussian-start fractions in `check_basins.py` part C: starts of scale σ below about u_s almost never learn, and $u_s \approx 2\gamma/s$ grows as the signal weakens, so a fixed init scale that works for $s = 1$ fails for $s = 0.4$. Ziyin et al.'s bound (16) puts the nonzero solutions at radius at least about $(\gamma/\|\mathbb{E}[xy]\|)^{1/(D-1)}/d_0$, which for $D = 2$, $d_0 = 1$ is the same γ/s scaling; Kaiming's $1/\sqrt{d_0}$ radius sits inside it when the signal is weak (their Figure 3).